A-A-433D
January 8, 1988
SUPERSEDING
A-A-433C
April 1, 1983

COMMERCIAL ITEM DESCRIPTION

CLAMP, C (HEAVY SERVICE)

The General Services Administration has authorized the use of this commercial item description in lieu of Type I, Class D, Federal Specification GGG-C-406.

Scope: This description establishes requirements for a C-clamp (heavy service) used for clamping and holding work in a fixed position.

Salient characteristics:

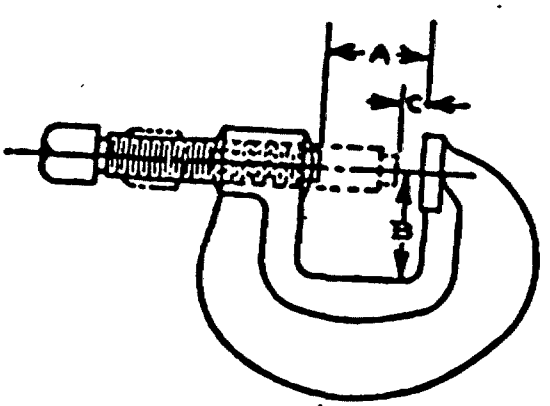
Material. Materials used in the manufacture of clamps shall be of uniform quality and free from defects and imperfections that may affect their serviceability or durability.

Frame. Steel, heat treated or malleable iron, designed to withstand minimum loads specified without evidence of damage or permanent deformation. The anvil of the frame shall have a smooth flat surface perpendicular to the center line of the class 2B, or better, right-hand threaded hole.

Screw. The screw shall be made of steel having a minimum tensile strength of 125,000 p.s.i. and be hardened to Rockwell C 22 minimum. The outer end of the screw shall have square wrenching surface with an across the flats dimension not less than the major diameter of the screw. The inner end of the screw shall have a cylindrical portion with a diameter approximately equal to the minor diameter of the threaded portion of the screw.

Finish. All surfaces of the clamps shall have a smooth finish with all edges rounded or beveled and shall be thoroughly cleaned and free of all foreign matter, fins, scale, flux, or any other imperfection.

Illustration. The clamps shall be similar to the illustration shown below.



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FSC 5120

National Stock Number	Size of clamp	A Full opening (min.)	C Closed opening (max.)	B Working depth (min.)	Diameter of screw (min.)	Weight of clamp (max.)	Test load (min.)
		Inches	Inches	Inch	Inch	Pounds	Pounds
5120-00-249-7491	3/4	3/4	0	9/16	5/16	1/2	2,500
5120-00-240-5618	1	1	0	15/16	3/8	3/4	5,600
5120-00-255-1117	1-1/2	7-1/2	3/4	1-7/16	7/16	1-3/4	8,750
5120-00-241-2973	2	2	7/8	1-11/16	9/16	3-1/2	12,500
5120-00-249-7492	3	3	1-3/8	2-3/16	9/16	6-1/2	16,250
5120-00-529-3744	4-1/2	4-1/2	2	2-11/16	7/8	10	20,000
5120-00-293-0147	5	5	2-1/2	2-15/16	7/8	13-1/2	23,800
5120-00-180-0897	6	6	3-1/8	3	1	18-1/2	27,500
5120-00-249-7494	8	8	4-1/4	3-7/16	1-1/8	26-1/2	31,250
5120-00-268-3521	10	10	6-3/8	3-13/16	1-1/8	34	35,000
5120-00-268-3522	12	12	7-5/8	3-15/16	1-1/8	43-1/2	40,000

Identification marking. Each clamp shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol, part number and country of origin.

Workmanship. The clamps shall be free of any defects that can adversely affect their operation, serviceability, or life expectancy.

QUALITY ASSURANCE PROVISIONS

Examination of the end item for defects in dimensions, finish, markings, and workmanship. Sampling shall be in accordance with MIL-STD-105 at inspection level II with an acceptable quality level (AQL) of 2.5.

Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105 at inspection level S-3 with an Acceptable Quality Level (AQL) of 4.0.

Tests:

Proof load test. Shall be conducted on a tension machine or similar device of known accuracy. Clamp shall be subjected to the applicable specified test load. The test shall be conducted with the clamp in the full open or minimum capacity opening as applicable. The test load shall be applied gradually, to simulate actual usage, until the specified minimum load is obtained. This load should be maintained for a period of not less than 5 minutes. Upon completion of this test, the clamp shall be examined and any evidence of deformed or damaged components shall be sufficient cause for classifying that clamp as a reject. The test load shall be applied to the center line of the screw.

Tensile test. The manufacturer shall furnish certification that the material of the screws conform to minimum tensile strength specified.

Hardness test. Clamp shall be tested for the hardness specified for the screws. Test impressions may be taken at any location near the extreme end of the threaded portion.

Regulatory requirement. In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking: The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

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Notes:

The issue of referenced documents in effect on the date of invitation for bid or request for proposals shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the Contracting Officer.

Military Standards are available from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians:

Army - GL
Air Force - 99

Review Activities:

Air Force - 84

PREPARING ACTIVITY

GSA-FSS

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